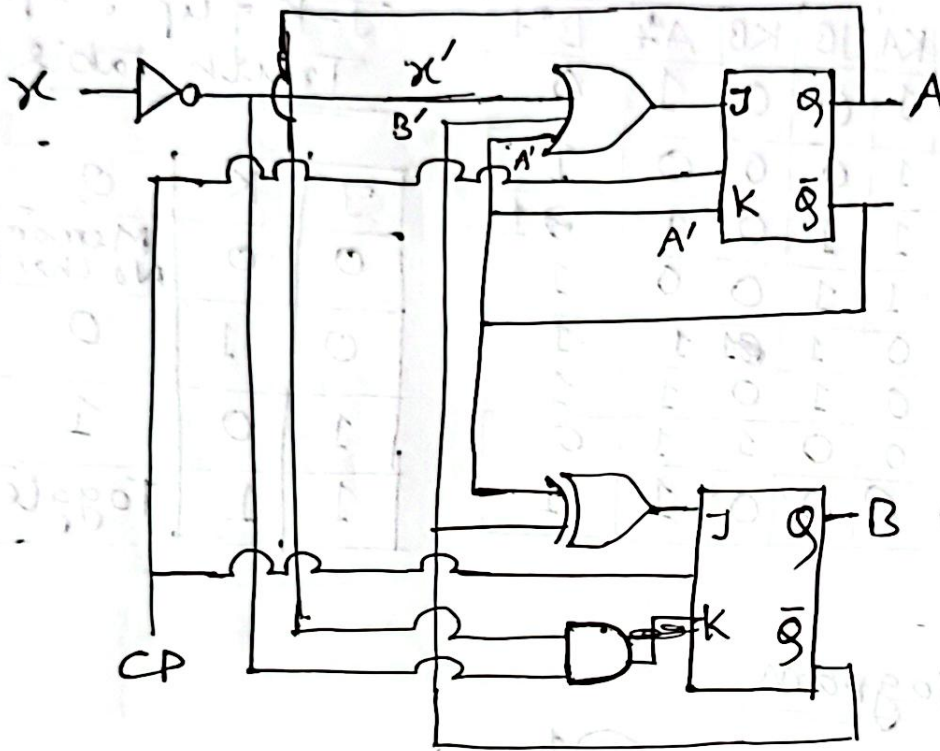


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$$J_A = A' + B' + x'$$

$$K_A = A'$$

$$J_B = A' \oplus B' = (A')'B' + A'(B')' = A \oplus B$$

$$K_B = Ax'$$

State Table

A	B	x	JA	JB KA	KB JB	KB	A ⁺	B ⁺
0	0	0	1	1	0	0	1	0
0	0	1	1	1	0	0	1	0
0	1	0	1	1	1	0	1	1
0	1	1	1	1	1	0	1	1
1	0	0	1	0	1	1	1	1
1	0	1	1	0	1	0	1	1
1	1	0	1	0	0	1	1	0
1	1	1	0	0	0	0	1	1

J-K flip flop truth Table

J	K	Q
0	0	Memory/ No change
0	1	0
1	0	1
1	1	Toggle

State Diagram

